

When Firms Can Opt Out: Minimum Wages in Swiss Collective Bargaining.

Evidence from Switzerland's Machinery, Electrical Engineering and Metals Industry.

Katja Hager
University of Zurich

March 13, 2026

Why should we talk about (voluntary) minimum wages?

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Little causal evidence in the Swiss context.

- ▶ Cantonal MW in Neuchâtel (Berger & Lanz, 2020).
- ▶ Master's theses on CBA-included MW
 - ▶ Suspension of generally binding CBA in hairdressing industry (Roth et al., 2024).
 - ▶ General bindingness of CBA in gastronomy sector (Gubler, 2025).

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Why do firms join CBAs, and do *voluntary* CBAs affect wages and employment?

This paper

This paper analyzes the introduction of minimum wages into the sectoral CBA in the MEM industry in 2013, which is voluntary and allows firms to opt out.

- ▶ How does the introduction of minimum wages impact wages and employment?
- ▶ Do firms opt out to evade wage floors?

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- ▶ How does the introduction of minimum wages impact wages and employment?
- ▶ Do firms opt out to evade wage floors?

Methodology

- ▶ Qualitative interviews with Social Partners
- ▶ Empirical analysis of wage survey data (ESS) using several empirical designs

Voluntary CBA membership

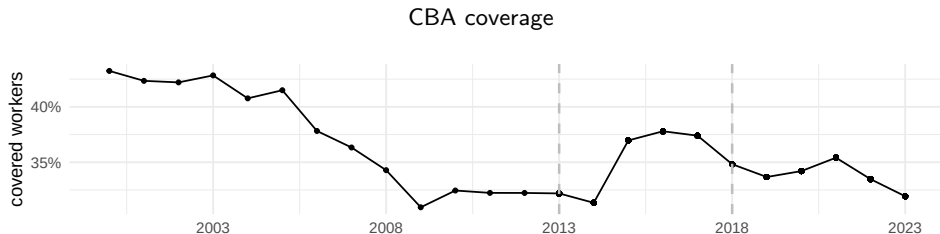
Swissmem (2025): “Swissmem member companies are absolutely free to choose whether or not to apply the CEA.”

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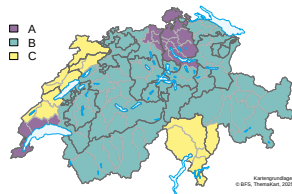


Source: Job Statistics (JOBSTAT) filtered by NACE codes, KOF data.

Minimum wage policy

CBA in effect from July 1, 2013 – June 30, 2018.

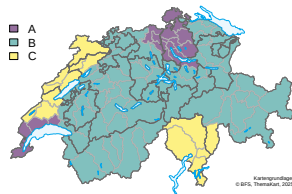
Monthly MW ($\times 13$)	
Region A	CHF 3,850.–
Region B	CHF 3,600.–
Region C	CHF 3,300.–



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- ▶ Long transition periods until mid-2016 or end of CBA.
- ▶ Low bite: 1.83% of CBA-covered workers, 5.02% among non-covered workers.
- ▶ Wage levels were determined to respect existing wage structures.

Measuring wage and employment effects: firm-level analysis

Swiss Earnings Structure Survey by FSO

- ▶ Sample period: 2006 - 2018
- ▶ Keep MEM sector, identified with NACE codes (80 sub-sectors)
- ▶ Keep workers 19-65 years old with employment rate 5% - 150%

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Firm-level outcomes

- ▶ Average monthly wage of bottom-half earners, hours adjusted
- ▶ Within-firm wage percentiles (hours adjusted)
- ▶ Within-firm change in employment

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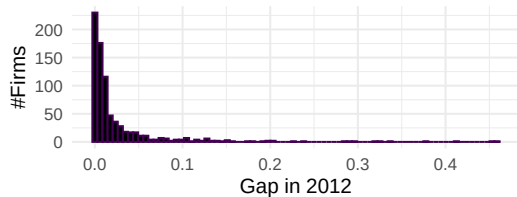
Empirical design

- ▶ Firm-level gap design similar to Derenoncourt and Weil (2025)
- ▶ Firm's shortfall between wage floor and 2012 wages as treatment intensity.
- ▶ All firms in MEM sector.

Empirical design: Firm-level gap

The gap measures a firm's exposure to the new minimum wages.

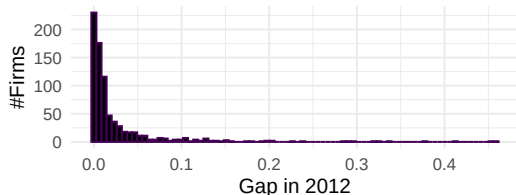
$$\text{GAP}_{f,2012} = \sum_{i \in f} \frac{\max\{MW_i - w_i, 0\}}{w_i}$$



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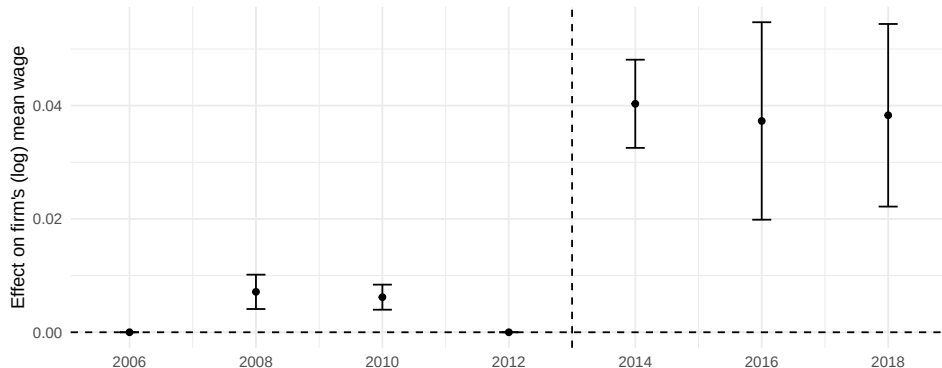


Use the standardized gap as continuous treatment variable.

$$\log w_{f,t} = \alpha + \sum_{k \neq 2012} \beta_k \text{GAP}_f \times \mathbb{1}_{[t=k]} + \zeta_f + \delta_t + \rho_r + \varepsilon_{f,t}$$

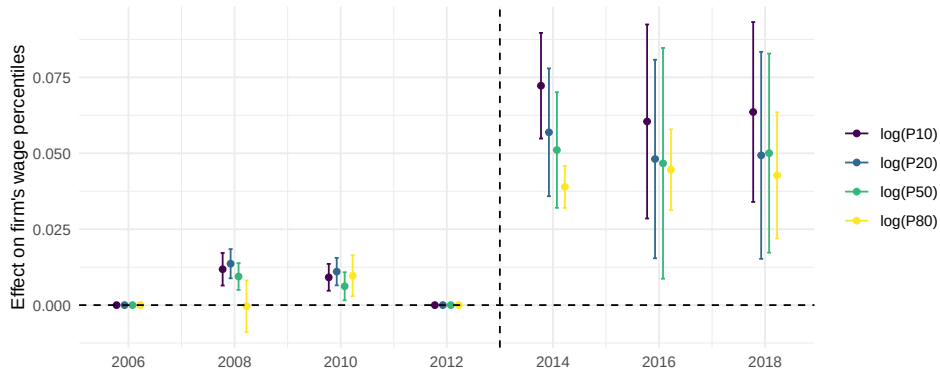
- ▶ Coefficient of interest: β_k
- ▶ Compare firms within MEM sector which had different wage structure in 2012.

Positive effect of minimum wage on (log) average firm wage



A coefficient of 0.04 in 2014 implies that firms with gaps one SD larger (original SD: $0.048 = 4.8$ pp) increase wages (below the median) by 0.04 log points $\approx 4\%$ more than less exposed.

Effect of minimum wage on wage percentiles



A coefficient of 0.075 on P10 implies that firms with gaps one SD larger (original SD: 4.84 pp) increase their tenth wage percentile by $\approx 7.5\%$ more than less exposed firms.

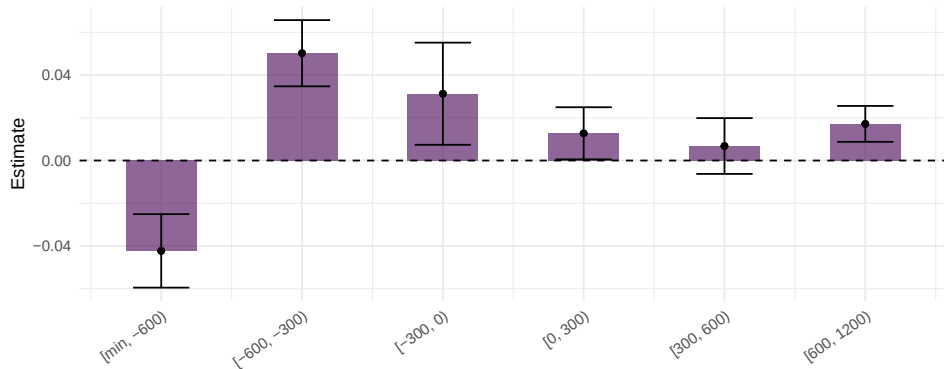
CBA triple interaction effect on wages

Dependent Variable:	log(mean wage)
Gap $\times$ Year = 2014	0.034*** (0.003)
Gap $\times$ Year = 2016	0.030*** (0.003)
Gap $\times$ Year = 2018	0.031*** (0.005)
Gap $\times$ CBA-covered $\times$ Year = 2014	0.017** (0.005)
Gap $\times$ CBA-covered $\times$ Year = 2016	0.058 (0.041)
Gap $\times$ CBA-covered $\times$ Year = 2018	0.039 (0.027)
Firm-fixed effects	Yes
Labour Market FE	Yes
Year FE	Yes
Observations	25,714

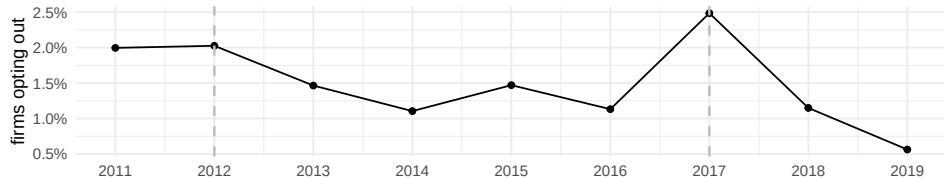
Clustered (Sub-sector & Greater Region) standard-errors in parentheses.

Effect of minimum wage on wage distribution

- ▶ Outcome = firm's employment share by wage bin $k = \frac{E_{f,t,k}}{E_{f,t}}$
- ▶ Aggregate policy effect for 2014-2018 relative to 2012



Opting-out as a way out?



Reading example: In 2014, 1.1% of CBA-covered firms opted out per next year.

Commitment to social partnership.

No opt-outs despite wage effects.

- ▶ Historical anchoring: “culture of social partnership” (Kareen Vaisbrot)
- ▶ Signal to attract workers
- ▶ Peace obligation
- ▶ Public procurement
- ▶ Flexibility in working hours

Source: Interviews with Kareen Vaisbrot (Swissmem) and Matteo Pronzini (Unia), press releases.

Conclusion and implications

Despite

- ▶ partial coverage and low bite of minimum wages
- ▶ MEM with small share of low-wage workers and high export-orientation
- ▶ difficult negotiations due to minimum wage introduction

We see that

- ▶ Voluntary MW increase wages without significantly impacting employment.
- ▶ No opting out to evade wage floors.

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Therefore

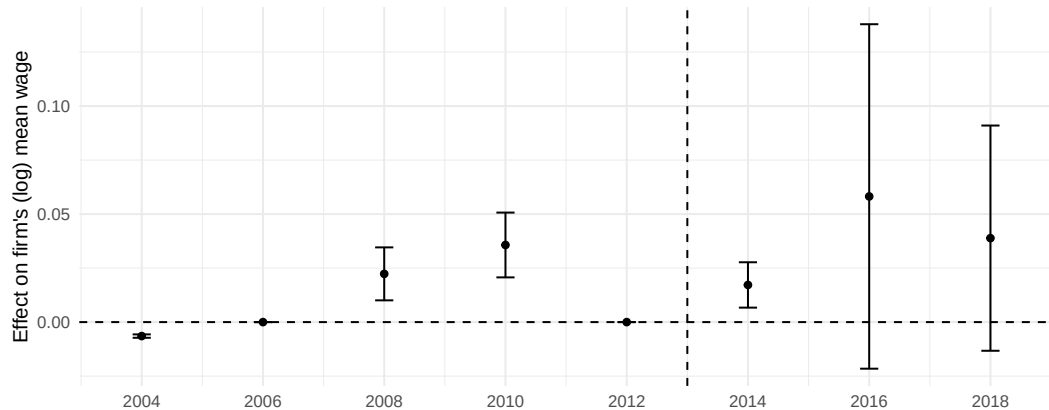
- ▶ Even minimum wages in voluntary CBAs can reduce wage inequality.
- ▶ Considerable spillovers to non-covered firms show the broader influence of sectoral agreements.
 - ▶ Expanding coverage could extend gains to additional groups.

Questions?

Appendix

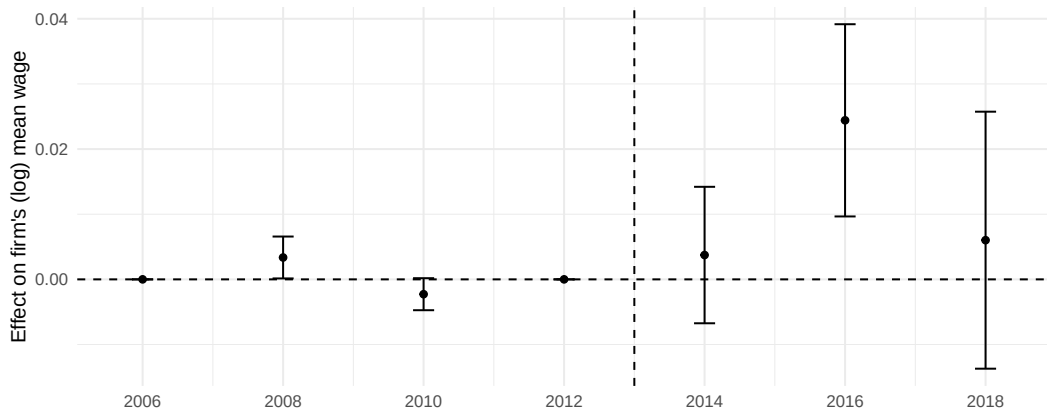
Differential treatment effects

Interaction of gap with CBA coverage.



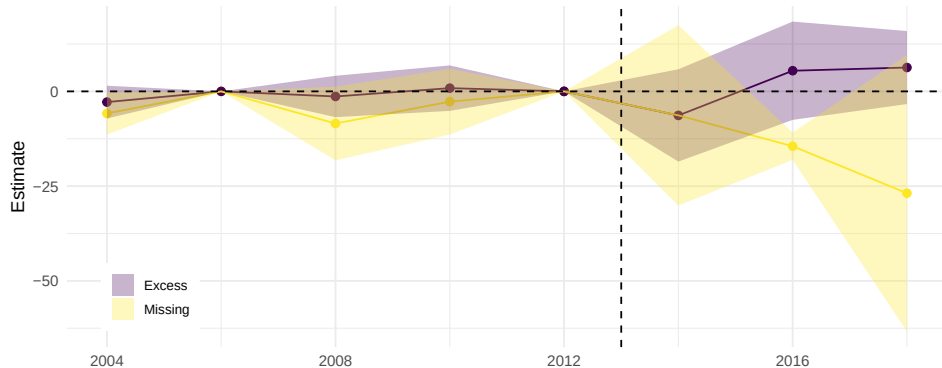
Differential treatment effects

Interaction of gap with transition period ending in 2016.



Effect of minimum wage on jobs

- Outcome = employment below and above MW



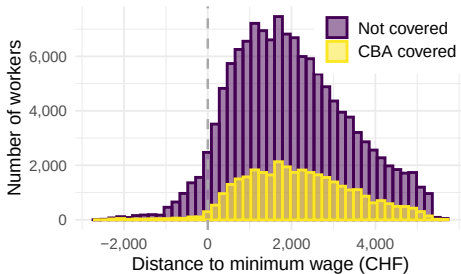
Drawing on international literature

- ▶ Germany: Introduction of national minimum wage in 2015 raised wages at the bottom, with limited spillovers to higher percentiles and no negative employment effects (Dustmann et al., 2021).
- ▶ Hungary: Increase in MW in 2001 leads to significant bunching at MW (high compliance) and few destroyed jobs (worker-level) ((Harasztosi & Lindner, 2019)).
- ▶ US: Introduction of voluntary MW by US retailers increases wages, no spillovers (Derenoncourt & Weil, 2025)
- ▶ South Africa: CBAs increase wage levels and induce large wage spillovers to non-covered firms (Bassier, 2022).
- ▶ UK: National minimum wage leads to spillovers within, not between firms (Giupponi & Machin, 2022).

Caveats with ESS data

- ▶ Survey data
 - ▶ Variable on CBA coverage is noisy and understates true coverage.
 - ▶ Survey data can be characterized by imprecise wage reporting.
- ▶ Repeated cross-section.
 - ▶ Substantial sample loss over time
 - ▶ We disproportionately lose small firms over time. Roughly 3/4 of CBA-covered minimum-wage workers are employed in small firms.
 - ▶ No clear compositional shift for workers.
 - ▶ Not a true panel.
- ▶ Structural break in 2012
- ▶ MEM-CBA and NACE codes do not fully align.
- ▶ Time period / unstable macroeconomic environment

Do the minimum wages (have) bite?



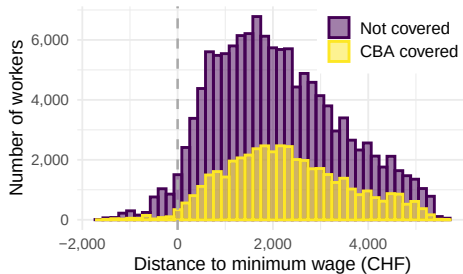
(a) Distance to minimum wage - counts



(b) Distance to minimum wage - density

Source: ESS.

How is compliance in 2018?



(a) Histogram



(b) Density

Source: ESS.

Minimum wage policy

- ▶ Partial coverage (ca. 35%)
- ▶ Low bite (1.83% of CBA-covered workers, 5.02% among non-covered workers).
- ▶ High compliance (1.42% among CBA-covered workers earn below the minimum, 3.1% among non-covered workers).
- ▶ Long transition period (mid-2016).

Political context of introduction

Union's demand for minimum wages in 2013

- ▶ Core issue: the principle of including collective wage negotiations.
- ▶ Political environment: proposals for a national minimum wage, wage caps (1:12 initiative), inheritance tax.
- ▶ Minimum wages were designed to respect existing wage structures.

Alternative designs

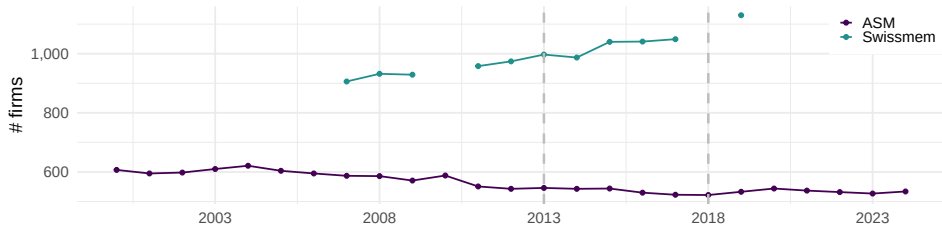
- ▶ Sub-sectoral analysis (aggregate)
 - ▶ results point in same direction
 - ▶ including firm fixed effects for unobserved firm heterogeneity is important (pre-policy effects)
 - ▶ wage growth is better as outcome variable
- ▶ DID analysis with the watch sector
 - ▶ export orientation, historical importance, collective bargaining, average tenure, age, and limited part-time employment, sectors have no overlapping sub-industries.
 - ▶ parallel trends assumption is hard to argue; The geographic clustering of the watch industry and its distinct work practices, combined with CBA changes at the same time, make it an inadequate counterfactual for MEM minimum wage effects.

MEM sector: Economically and politically important.

- ▶ Largest industrial employer (330'000 workers, 10'000 establishments)
- ▶ 5% of Swiss GDP
- ▶ Export-orientation
- ▶ Influential for national politics and other CBAs
- ▶ Historical sector

ASM - Swissmem membership

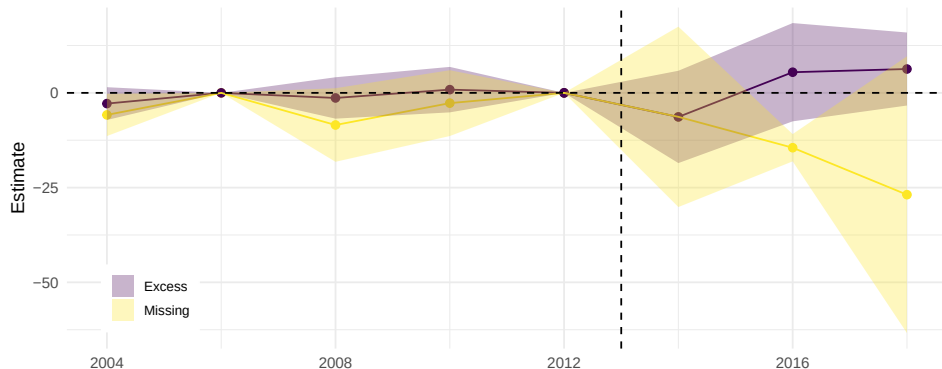
- ▶ Contracting partner: Arbeitgeberverband der Schweizer Maschinenindustrie (ASM)



Source: Annual reports of Swissmem.

Effect of minimum wage on jobs below MW & up to CHF 600 above

- ▶ Outcome = firm's change in employment w.r.t. to pre-policy
- ▶ $employment_f$ = sum of full-time workers in those bins



Effect of minimum wage on within-firm employment

